Def. Let K be a Commutative Ring with identity.
A map $D : M_{n\times n}(K) \to K$ is called n-linear if: for each $1 \le i \le n$,

$$D\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ c\alpha_i + \alpha_i' \\ \vdots \\ \alpha_n \end{pmatrix} = c \cdot D\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_i \\ \vdots \\ \alpha_n \end{pmatrix} + D\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_i' \\ \vdots \\ \alpha_n \end{pmatrix}$$

when $c \in K$, $\alpha_1, \cdots, \alpha_n, \alpha_i'$ are arbitrary row vectors.

Def. A n-linear map $D : M_{n\times n}(K) \to K$ is called Alternating if:

$D(A) = 0$ whenever two rows of A are equal.

Lemma. If n-linear map $D : M_{n\times n}(K) \to K$ is Alternating,
then for A', obtained from A by interchanging two rows of A, $D(A') = -D(A)$.
Prof sketch. For adja

Def. Let K be a commutative Ring with identity.
The unique map $D: M_{n\times n}(K) \to K$ satisfying

n-linear, alternating, $D(I)=1$

is called the determinant function.

Example. The determinant of 2 by 2 matrix is determined by: $\det A = A_{11}A_{22} - A_{12}A_{21}$.
Given $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = (A_{11}e_1 + A_{12}e_2,\ A_{21}e_1 + A_{22}e_2)$

Since D is n-linear,

$$D(A) = D(A_{11}e_1 + A_{12}e_2,\ A_{21}e_1 + A_{22}e_2)$$
$$= A_{11}D(e_1, A_{21}e_1 + A_{22}e_2) + A_{12}D(e_2, A_{21}e_1 + A_{22}e_2)$$
$$= A_{11}A_{21}D(e_1,e_1) + A_{11}A_{22}D(e_1,e_2) + A_{12}A_{21}D(e_2,e_1) + A_{12}A_{22}D(e_2,e_2)$$

Since D is alternating, $D(e_1,e_1) = D(e_2,e_2) = 0$ and $D(e_2,e_1) = -D(e_1,e_2) = -D(I) = -1$.
So,
$$= A_{11}A_{22} - A_{12}A_{21}$$ $\boxtimes$

A Matrix $\operatorname{adj} A = \begin{bmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{bmatrix}$ is called the classical adjoint of A.
Using the properties of det, n-linear, alternating, $\det I = I$,
deduce the following:
a). $(\operatorname{adj} A)\cdot A = A\cdot(\operatorname{adj} A) = (\det A)\cdot I$.
b). $\det(\operatorname{adj} A) = \det A$
c) $\operatorname{adj}(A^t) = (\operatorname{adj} A)^t$.
Proof. The first claim can be shown directly: $(\operatorname{adj} A)\cdot A = \begin{bmatrix} A_{22}A_{11} - A_{12}A_{21} & 0 \\ 0 & -A_{21}A_{12} + A_{11}A_{22} \end{bmatrix}$
Similarly b) and c).

Given a $n \times n$ matrix A over a commutative Ring with identity K, the i'th row α_i is for $1 \le i \le n$,

$$\alpha_i = \sum_{j=1}^{n} A_{ij} \cdot e_j .$$

Therefore, for n-linear, alternating function D,

$$D(A) = D(\alpha_1, \alpha_2, \cdots, \alpha_n) = \sum_{j=1}^{n} A_{1,j} \cdot D(e_j, \alpha_2, \cdots, \alpha_n)$$

$$= \sum_{j_1=1}^{n} A_{1,j_1} \cdot \sum_{j_2=1}^{n} A_{2,j_2} \cdot D(e_{j_1}, e_{j_2}, \alpha_3, \cdots, \alpha_n)$$

$$\vdots$$

$$= \sum_{1 \le j_1, \cdots, j_n \le n} A_{1 j_1} \cdot A_{2 j_2} \cdot \cdots \cdot A_{n j_n} \cdot D(e_{j_1}, \cdots, e_{j_n})$$

Since D is alternating, if there are two of same indices, then $D(e_{j_1}, \cdots, e_{j_n}) = 0$. So, we can consider $(e_{j_1}, \cdots, e_{j_n})$ as a permutation on n letters, moreover, if $D(I) = 1$, then $D(e_{j_1}, \cdots, e_{j_n}) = \operatorname{sgn}(j_1, \cdots, j_n)$. In sum,

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn} \sigma \cdot A_{1,\sigma(1)} \cdot \cdots \cdot A_{n,\sigma(n)}.$$

「Invertibility of Matrix.

「Lemma. Let K be a Commutative Ring with identity, and let $A, B \in M_{n\times n}(K)$.
Then $\det AB = \det A \cdot \det B$.

Proof. Let B be a fixed $n\times n$ matrix over K. Define $D(A) = \det AB$.
Denote $\alpha_1, \cdots, \alpha_n$ as j'th rows of A. Then $D(\alpha_1, \cdots, \alpha_n) = \det(\alpha_1 B, \cdots, \alpha_n B)$.
Moreover, D is n-linear and alternating, since det is. So, $D(A) = \det A \cdot D(I) = \det A \cdot \det B$.

「Def. Given $A \in M_{n\times n}(K)$, define the Classical adjoint of A as a Matrix

$$(\operatorname{adj} A)_{ij} \overset{\text{def}}{=} (-1)^{i+j} \cdot \det A(j|i).$$

where $A(j|i)$ is a $(n-1)$ by $(n-1)$ Matrix obtained by remove j'th row and i'th Column.

「Thm. Let K be a Commutative Ring with identity.
Given $A \in M_{n\times n}(K)$,

$$(\operatorname{adj} A) \cdot A = (\det A) \cdot I.$$

Proof.